chemicals includes **fertilizers, liming and acidifying agents, soil conditioners, pesticides and chemicals** used in animal husbandry, such as antibiotics and hormones.

Excessive use of fertilizers and pesticides leads to the contamination of groundwater and makes it nonpotable, ultimately affecting the soil fertility. Most of the chemical fertilizers contain varying amounts of nitrogen, phosphorous, potassium and nutrients that plants need to grow. Soil acidity influences C and N cycles by affecting soil microbes, also green house gas flux in soils and affect bio availability of N, P, S like major nutrients. This makes the soil too acidic or alkaline so that it becomes difficult for the plants to survive. These residues and synthetic chemicals like DDT (dichloro diphenyl trichloro ethane) and PCBs (polychlorinated biphenyls) cause nutrient and pH imbalance and quality reduction of agricultural produce. This problem can be minimised by sustainable agriculture.

Pesticides increase incidence of brain, blood cancer and neurotoxicity, Parkinson like symptoms, infertility, birth defects, reproductive and behavioural disorders.



- Nitrates from fertilizers interact with the haemoglobin to form methyl haemoglobin. This reduces oxygen uptake, results in Blue baby syndrome (cyanosis) and hypoxia. Nitrates vasodilate and reduce blood pressure.
- **Bio-magnification:** Pollutants, toxic substances increase in water move from one food chain to many and finally reach human being and this process of bio-amplification or increase in concentration is called bio-magnification.

8.6 Alien invasive species

Invasion of alien or introduced species disrupts ecosystem processes, threaten biodiversity, reduce native herbs, thus reducing the ecosystem services (benefits). During eradication of these species, the chemicals used increases greenhouse gases. Slowly they alter ecosystem, micro climate and nature of soil and make it unsuitable for native species and create human health problems like allergy, thus resulting in local environmental degradation and loss of important local species.

According to World Conservation Union invasive alien species are the second most significant threat to bio-diversity after habitat loss.

What is invasive species?

A non-native species to the ecosystem or country under consideration that spreads naturally, interferes with the biology and existence of native species, poses a serious threat to the ecosystem and causes economic loss.

It is established that a number of invasive species are accidental introduction through ports via air or sea. Some research organisations import germplasm of wild varieties through which also it gets introduced. Alien species with edible fruits are usually spread by birds.

Invasive species are fast growing and are more adapted. They alter the soil system by changing litter quality thereby affecting the soil community, soil fauna and the ecosystem processes.

It has a negative impact on decomposition in the soils by causing stress to the neighbouring native species. Some of the alien species which cause environmental issues are discussed below

Eichhornia crassipes



Figure 8.5: *Eichhornia crassipes*

It is an invasive weed native to South America. It was introduced as aquatic ornamental plant, which grows faster throughout the year. Its widespread growth is

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